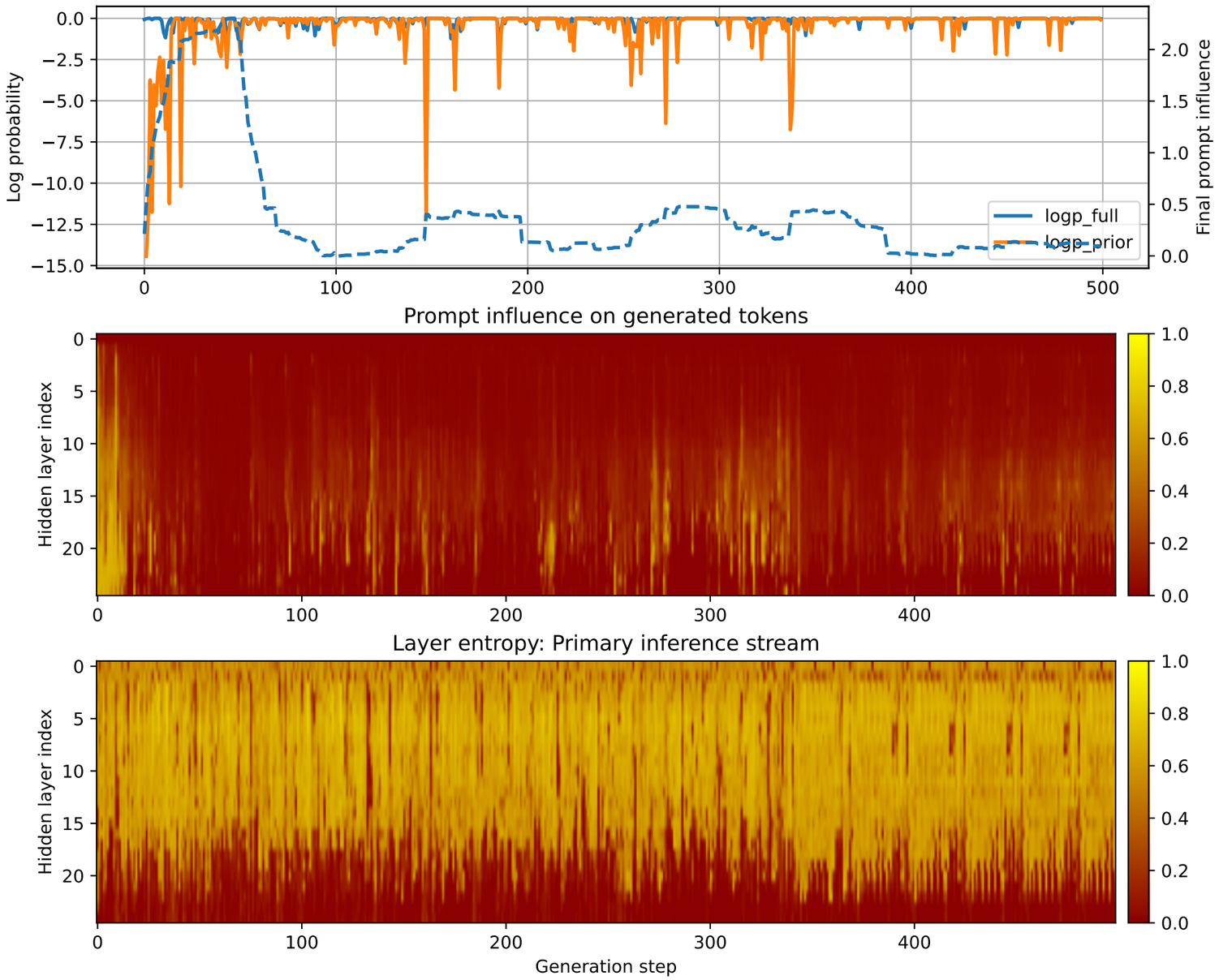
Prompt 1




```
PROMPT:
Task ID: PYC044  Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests.  Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC044  Your code must contain this required symbol:
def binary_search  Implement this task:
define function binary_search(xs,target) returning index or -1..
Start your answer with the check-token line,
then provide the Python implementation.

MODEL RESPONSE:
# CHECK_TOKEN: PYC044  # Define function binary_search def
binary_search(xs, target):     """     Returns the
index of the target value in the given list of integers, `xs`,
using a binary search algorithm.     The binary
search algorithm works by repeatedly dividing the search interval
in half until the target value is found or
the search interval is empty.      Args:         xs (list[int]):
A list of integers to search through.
target (int): The value to find in the list.      Returns:
 int: The index of the target value in the
list, or -1 if the target value is not found.     """     # Check
if the list is empty     if len(xs) == 0:
return -1      # Initialize the left and right pointers     left
= 0     right = len(xs) - 1      # Perform
binary search     while left <= right:         # Calculate the
middle index         mid = (left + right) // 2
# Check if the middle element is the target value         if
xs[mid] == target:             return mid
# If the target value is less than the middle element, update the
left pointer to be the middle + 1
elif xs[mid] > target:             right = mid - 1          # If
the target value is greater than the middle
element, update the right pointer to be the middle - 1
else:             left = mid + 1      # If the
target value is not found, return -1     return -1  # Test the
binary_search function print(binary_search([1,
2, 3, 4, 5], 3))  # Output: 2 print(binary_search([1, 2, 3, 4,
5], 6))  # Output: -1 print(binary_search([1,
2, 3, 4, 5], 7))  # Output: -1 print(binary_search([1, 2, 3, 4,
5], 8))  # Output: -1 print(binary_search([1,
2, 3, 4, 5], 9))  # Output: -1 print(binary_search([1, 2, 3, 4,
5],
```